

School of Science and Engineering

AI 501 Mathematics for Artificial Intelligence

ASSIGNMENT 1

Due Date: 1 pm, Sunday, October 12, 2025.

Format: 9 problems for a total of 100 marks

Instructions:

- You are allowed to collaborate with your peers but copying your colleague's solution is strictly prohibited. This is not a group assignment. Each student must submit his/her own assignment.
- Solve the assignment on blank A4 sheets and staple them before submitting.
- Submit in-class or in the dropbox labeled AI-501 outside the instructor's office.
- **Write your name and roll no. on the first page.**
- Feel free to contact the instructor or the teaching assistants if you have any concerns.

- You represent the most competent individuals in the country, do not let plagiarism come in between your learning. In case any instance of plagiarism is detected, the disciplinary case will be dealt with according to the university's rules and regulations.
- We require you to acknowledge any use or contributions from generative AI tools. Include the following statement to acknowledge the use of AI where applicable.

I have used [insert Tool Name] to [write, generate, plot or compute; explain specific use of generative AI] [number of times].

Problem 1 (10 marks)**Systems of Linear Equations and Null Spaces (Sensor Fusion in Robotics)**

A mobile robot is trying to localize itself in a 2D environment. Its state is described by three variables:

$$x_1 = \text{horizontal position}, \quad x_2 = \text{vertical position}, \quad x_3 = \text{orientation}.$$

Three different sensor subsystems provide the following constraints:

- Sensor 1 reports that $x_1 - 2x_2 + 3x_3 = 1$.
 - Sensor 2 reports that $-x_1 + 2x_2 - 4x_3 = -1$.
 - Sensor 3 reports that $2x_1 - 4x_2 + 6x_3 = 2$.
- (a) Construct the system of linear equations in matrix form $Ax = b$ from the above sensor constraints.
- (b) Solve the system $\mathcal{L}(A, b)$. *Hint: Use systematic row operations to eliminate variables and reduce the system step by step.*
- (c) Identify the free variables and determine the rank of A .
- (d) Give the full set of solutions.
- (e) Express the general solution as a sum of a particular solution w plus a vector u from $\text{Null}(A)$.

Problem 2 (10 marks)**Linear Independence, Singular Matrices, and Solutions of a SLE**

In data science and AI, we often represent information as vectors in a feature space. For example, suppose you are building a model where the columns of a matrix A represent different feature directions. If these features are linearly independent, then they provide unique and non-redundant information. If they are dependent, some features can be expressed as combinations of others, leading to redundancy.

Similarly, when we try to fit data (represented by a vector b) into this feature space, the system of linear equations $\mathcal{L}(A, b)$ corresponds to finding a combination of features that explains the data. Whether this system is consistent depends on whether b lies in the span of the columns of A .

- (a) Prove that if the column vectors of a square matrix A are linearly independent, then A is a non-singular (invertible) matrix.
- (b) Prove that the system of linear equations $\mathcal{L}(A, b)$ is consistent $\iff b$ belongs to the span of the columns of A .

Problem 3 (10 marks)**Vector Spaces, Subspaces, and Dimension**

In some physical or sensing systems the response to a scalar stimulus differs by direction: one component responds linearly (keeping sign), while another responds only to the magnitude of the stimulus (losing sign information). Consider the set $V = \mathbb{R}^2$ with the usual vector addition $+$ but with a modified scalar multiplication

$$\alpha * \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} \alpha x \\ |\alpha| y \end{pmatrix} \quad (\alpha \in \mathbb{R}, (x, y) \in V).$$

This models an anisotropic scaling that discards the sign of the scalar on the second coordinate.

- (a) Prove that $(V, +, *)$ satisfies all axioms of a vector space but one. State which one and provide an example.

In AI and robotics, polynomials are often used to approximate smooth trajectories or signals. For example, a cubic polynomial in $P_3(\mathbb{R})$ might describe the position of a robot arm over time, or approximate a nonlinear function in regression.

Now suppose we are interested in the set of cubic polynomials that satisfy two specific physical constraints:

$$W = \{p(x) \in P_3(\mathbb{R}) \mid p'(-1) = 0 \text{ and } p''(1) = 0\}.$$

Here:

$p'(-1) = 0$ means the trajectory has a horizontal slope at $x = -1$ (e.g., the robot is momentarily at rest).

$p''(1) = 0$ means the curvature at $x = 1$ is zero (e.g., the path is locally straight at that point).

- (b) Prove or disprove that W is a linear subspace of $P_3(\mathbb{R})$.
(c) If it is a subspace, find a basis of W .
(d) If it is a subspace, determine the dimension of W .

Problem 4 (10 marks)**Matrix Inverse**

Matrices represent transformations of data. For example: In computer graphics or robotics, a matrix A might transform coordinates from one frame to another. In machine learning, matrices are used to represent weight transformations or covariance structures.

Invertibility tells us whether a transformation can be reversed — i.e., whether we can recover the original data from the transformed data.

- (a) Suppose $A^2 = 0$. Prove that A is not invertible. (Interpretation: this corresponds to a transformation that, when applied twice, sends every vector to zero — a process that cannot be reversed.)
- (b) Prove that if A is invertible, then

$$(A^{-1})^T = (A^T)^{-1}.$$

(This identity is crucial in optimization, since transposed and inverse matrices frequently appear together in gradient updates and covariance analysis.)

Problem 5 (10 marks)**Gram-Schmidt**

We often want to simplify vector computations by working with orthogonal or orthonormal sets of vectors. For example, in Principal Component Analysis (PCA) and other dimensionality-reduction techniques, we want to re-express data in a basis where the directions are orthogonal. This makes projections easy to compute and interpretations cleaner.

However, most sets of vectors we encounter are not orthogonal to begin with. The Gram–Schmidt process helps us systematically transform a set of linearly independent vectors into an orthogonal (or orthonormal) set spanning the same subspace.

Consider the vectors

$$v_1 = \begin{pmatrix} -7 \\ -i \\ 0 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2i \\ 3 \\ -1 \\ i \end{pmatrix} \in \mathbb{C}^4.$$

- (a) In practice, data features are rarely orthogonal. Apply the Gram–Schmidt procedure to $\{v_1, v_2\}$ to produce an orthogonal set. (This demonstrates how we can start with arbitrary features and produce ones that are easier to use for projections and decompositions.)
- (b) In high-dimensional spaces, we often want a full orthonormal basis to represent any possible vector. After orthogonalizing v_1 and v_2 , determine whether it is possible to extend your orthonormal set to a full basis for $\mathbb{C}^4$. If so, construct such a basis. (This connects to building complete coordinate systems, e.g. for QR factorization in machine learning.)
- (c) Subspaces defined by orthogonality conditions play a central role in machine learning, for example in defining null spaces and constraint spaces. Consider

$$W = \{v \in \mathbb{C}^4 \mid \langle v, v_1 \rangle = 0\}.$$

Determine whether W is a subspace of $\mathbb{C}^4$. (This shows how orthogonality naturally defines whole subspaces of valid solutions or constraints.)

Problem 6 (10 marks)**Chebyshev Inequality (Bounding Outliers in Data)**

In machine learning and data analysis, we often work with high-dimensional vectors representing samples, features, or signals. A useful way to summarize the overall magnitude of a vector is its *root mean square (RMS) value*, which measures the average energy of the components.

The Chebyshev inequality gives us a way to bound the number of “large” entries in a vector: if the RMS is small, not many entries can be very large in magnitude. This is directly connected to detecting outliers in datasets and understanding variance in features.

- (a) Suppose x is a 100-dimensional vector with $\text{rms}(x) = 1$. What is the maximum number of entries of x that can satisfy $|x_i| \geq 3$?

If your answer is k , explain why no such vector can have $k + 1$ entries with absolute values at least 3. Then construct an explicit example of a 100-vector with RMS value 1 that achieves exactly k entries larger than 3 in absolute value.

(Interpretation: This part asks you to quantify how many extreme outliers are even possible, given the average energy of the vector.)

- (b) Show that at least one entry of a vector must have absolute value at least as large as the RMS value of the vector.

(Interpretation: This guarantees the existence of at least one entry that is “representative” of the overall energy — no vector can have every entry below the RMS in magnitude.)

Problem 7 (10 marks)

Linear, Affine and Convex Combination We require you to plot the linear, affine, and convex combinations of vectors in $\mathbf{R}^3$. Write a Python script to generate three random vectors v_1, v_2, v_3 , where each coordinate of these vectors is drawn from a standard normal distribution. We require you to submit both the code and the resulting plots.

Problem 8 (15 marks)**Regularized Demand Forecasting (Ridge Regression & Train/Test Evaluation)**

An online retailer wants a stable linear model to forecast daily revenue from two marketing signals: number of site visitors (x_1) and paid-ad spend (x_2). Because these two signals are highly correlated in practice, ordinary least squares (OLS) can give large, unstable coefficients. Ridge regression (an L_2 penalty) is commonly used to stabilize the estimates and reduce variance at the cost of bias. Evaluating model performance on a held-out test set helps show whether regularization improves predictive performance in practice.

You are given the dataset below (each row is an observation):

obs	x_1 (visitors)	x_2 (ad spend)	y (revenue)
1	10	2	25
2	12	1	28
3	20	3	45
4	22	4	48
5	30	3	70
6	35	5	82

Let the linear model be

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

and define the design matrix $A \in \mathbb{R}^{6 \times 3}$ containing a column of ones (intercept) and the two feature columns.

- (a) **(Construct & OLS).** Construct the design matrix A and the response vector y . Write the OLS closed-form for $\hat{\beta}_{\text{OLS}} = (A^\top A)^{-1} A^\top y$. (Show the shapes of matrices involved.) Compute $\hat{\beta}_{\text{OLS}}$ numerically from the given data.

Why: this forces students to translate data into matrix form and perform the baseline estimation.

- (b) **(Ridge solution & numerical comparison).** For ridge penalty parameter $\lambda = 10$, compute the ridge estimator

$$\hat{\beta}_{\text{ridge}} = (A^\top A + \lambda I_3)^{-1} A^\top y.$$

(Note: do *not* regularize the intercept if you prefer — clearly state which convention you use.) Compute $\hat{\beta}_{\text{ridge}}$ numerically.

Why: shows how regularization changes estimates when predictors are correlated.

- (c) **(Train / Test MSE).** Use the first 4 observations (rows 1–4) as the training set and the final 2 observations (rows 5–6) as the test set. For both $\hat{\beta}_{\text{OLS}}$ and $\hat{\beta}_{\text{ridge}}$:

- compute the training mean squared error (MSE) and the test MSE,
- report which estimator gives lower test MSE and explain why (in terms of bias/variance and the effect of collinearity).

Why: demonstrates the practical value of regularization for generalization.

- (d) **(Conditioning and interpretation).** Compute the condition number of $A^\top A$ (ratio of largest to smallest eigenvalue). Explain briefly (2–3 sentences) how a large condition number justifies using ridge regression in this problem.

Why: connects algebraic properties (conditioning) to numerical instability and regularization.

Problem 9 (15 marks)

A hospital deploys a logistic regression model to estimate the probability that incoming patients have a serious condition (“positive”). Clinical decisions have asymmetric costs: missing a true positive (false negative) is typically much more costly than a false alarm (false positive). Thus, choosing a decision threshold matters; so does understanding precision/recall tradeoffs and the ROC/AUC.

You are given predicted probabilities from a trained logistic model and the true labels for 10 patients:

patient	1	2	3	4	5	6	7	8	9	10
$\hat{p}$	0.95	0.89	0.76	0.65	0.55	0.45	0.35	0.25	0.15	0.05
y	1	1	1	0	1	0	0	1	0	0

(There are five true positives and five true negatives.)

- (a) **(Threshold 0.5 — metrics).** Using threshold $t = 0.5$ (predict positive when $\hat{p} \geq 0.5$), compute:

- the confusion matrix (TP, FP, FN, TN),
- accuracy, precision, recall (sensitivity), specificity, and F1 score.

Why: shows how thresholding maps probabilities to discrete decisions and how different metrics measure different errors.

- (b) **(Cost-sensitive decision).** Suppose the clinical cost of a false negative is $C_{\text{FN}} = 10$ and the cost of a false positive is $C_{\text{FP}} = 1$ (correct predictions cost 0).

- Derive the optimal decision rule (threshold in terms of C_{FP} and C_{FN}) that minimizes expected cost for a single patient given predicted probability $\hat{p}$.
- Using the derived threshold, classify the 10 patients and compute the resulting confusion matrix and total expected cost (sum of realized misclassification costs).

Why: connects probabilistic outputs to operational decisions under asymmetric costs.

- (c) **(ROC & AUC).** Construct the ROC curve points by sweeping the threshold through the distinct predicted probabilities (include thresholds just above each distinct $\hat{p}$ and just below), compute the true positive rate (TPR) and false positive rate (FPR) at each step, and approximate the Area Under the Curve (AUC) using the trapezoidal rule. Report the AUC numeric value.

Why: AUC summarizes rank quality of the probabilistic model across thresholds.

- (d) **(Logistic formulation).** For a general binary dataset $\{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$ and logistic model $P(y_i = 1 \mid x_i) = \sigma(w^\top x_i)$ where $\sigma(s) = 1/(1 + e^{-s})$:

- write the negative log-likelihood $L(w)$ (sum over examples),
- derive the gradient $\nabla L(w)$ and the Hessian $H(w)$ in matrix form,

Why: connects the loss function used to train logistic regression with the practical second-order optimization algorithm commonly taught for numerical fitting.

— End of Assignment —